TEACHER COPY · ANSWERS

Bridge to Year 9 Advanced

Year 8 Advanced · 7 × 50-minute lessons · Mandelbrot series

Overview

BOOKLET PAGES	SECTION	CORE OUTCOMES	EXTENSION (GOLD / ★)
3–4	Pre-test (20 marks, 20 min)	One question per lesson; use it to decide where each student starts.	
5–7	Lesson 1: Algebra basics	MA4-ALG-C-01	Gold: MA5-ALG-C-01 (algebraic fractions with numerical denominators) · MA5-ALG-P-01/P-02 (indices, binomial numerators)
8–10	Lesson 2: Expanding brackets	MA4-ALG-C-01	★ negative coefficients: MA5-ALG-C-01 · Gold: binomial products MA5-ALG-C-01; special products MA5-ALG-P-02
11–13	Lesson 3: Factorising	MA4-ALG-C-01	★ negative HCF: MA5-ALG-P-01 · Gold: binomial common factors, difference of squares, monic trinomials MA5-ALG-P-01, MA5-ALG-P-02
14–16	Lesson 4: Indices	MA4-IND-C-01	Gold: index laws with pronumerals and the negative index MA5-IND-C-01
17–19	Lesson 5: Square roots & Pythagoras' theorem	MA4-IND-C-01 (roots) · MA4-PYT-C-01	Gold: surds MA5-IND-P-02 · length of an interval MA5-LIN-C-01 · 3D Pythagoras MA5-TRG-P-01
20–22	Lesson 6: Equations & formulas	MA4-EQU-C-01	Gold: 3-step equations and grouping symbols MA5-EQU-C-01 · two fractions, changing the subject MA5-EQU-P-02
23–25	Lesson 7: Area & volume	MA4-ARE-C-01 · MA4-VOL-C-01	Gold: composite solids and algebraic area/volume (Stage 4 content, harder problems)
26–27	Post-test (20 marks, 20 min)	Parallel form of the pre-test.	

Notes for the teacher

- **Levels.** Bronze = support (nearly everyone unaided), Silver = core Stage 4, Gold = extension (Stage 5 core or Path content, labelled with codes above). MAO-WM-01 applies throughout.
- **★ items.** The NCM 8 Advanced textbook teaches some content in Year 8 that the 2022 syllabus places in Stage 5: expanding with negative coefficients (MA5-ALG-C-01) and factorising with a negative HCF (MA5-ALG-P-01). These appear in Silver marked ★. Test Q2b and Q3c also use them.
- **Interleaving.** 61 of the 308 drill questions (20%; 15%–36% per lesson) come from an earlier lesson (chip ↻ L1–L6) or from Year 8 topics not taught in the booklet (integers, fractions, percentages, ratios, linear relationships, equations).
- **Timing.** 348 questions in total (41–55 per lesson). In 50 minutes most students will finish Bronze and most of Silver; Gold and any leftovers suit early finishers or homework.
- **No calculator.** Numbers are chosen so that every question can be done by hand; non-square roots are left exact (e.g. $\sqrt{74}$) and estimated to 1 d.p. by squaring.
- **Every answer below was checked by computer algebra** (expansions, factorisations, solutions, substitutions and all numerical values).

1 Algebra basics

B Bronze round

1	$10a$	**2**	$5m$	**3**	$8x + 2y$	**4**	$5p + 2q$
5	$12k$	**6**	$10ab$	**7**	$3m^2$	**8**	$4y$
9	$4a$	**10**	5 ↻ Integers	**11**	$6x + 4$	**12**	$-12n$
13	27 ↻ Fractions	**14**	$6ab$				
15	10	**16**	-8	**17**	20	**18**	5
19	20 ↻ Percentages	**20**	0				

S Silver round

21	$2x^2 + 7x$	**22**	$ab + 2a$	**23**	$12m^2n$	**24**	$6x^2y^2$

25 $-3p$ **26** $3a$ **27** $\frac{2x}{3}$ **28** $3:4$ ↻ Ratios

29 $8x$ **30** $2x$ **31** 4 **32** 6

33 $y=-11$ ↻ Linear **34** \$69 ↻ Percentages **35** -30 ↻ Integers

36 -1 **37** -9 **38** 4 **39** 2

40 a) $14n+6$ b) \$48

41 e.g. $x=2,\ y=1$: $3x+2y=8$ but $5xy=10$. $3x$ and $2y$ are unlike terms, so they cannot be added. (Note: $x=y=1$ gives $5=5$, which does *not* disprove it.)

G Gold round

42 $\frac{7x}{12}$ **43** $\frac{a}{15}$ **44** $\frac{7m}{12}$ **45** $\frac{2x}{3y}$

46 $\frac{7x}{12}$ **47** $6a$ **48** $\frac{3}{2}$ **49** $\frac{5x+3}{6}$

50 a) $2(3x+2)+2(x-1)=6x+4+2x-2=8x+2$ b) 34 c) the width $x-1$ must be positive

51 $56.25-37.5+6.25=25$; shortcut: it equals $(a-b)^2=5^2=25$

52 a) 290 b) Yes: $n=5$ gives $75-5=70$ (the 5th term)

53 58 (the numbers are 3 and 7: $9+49=58$; or $(a+b)^2-2ab=100-42$)

2 Expanding brackets

B Bronze round

1 $3x+12$ **2** $5a-10$ **3** $6m+2$ **4** $8-4y$

5 x^2+6x **6** $ab-3a$ **7** $3x+5$ ↻ L1 **8** $12p+18q$

9 24 ↻ Integers **10** $21-14k$ **11** $3x+20$ **12** 2 ↻ L1

13 $2x+11$ **14** $7a+4$ **15** $\frac{5}{6}$ ↻ Fractions **16** $4y-y^2$

17 $5x+8$ **18** 45 ↻ Percentages

S Silver round

19 $-2x-10$ **20** $-m-4$ **21** $-12a+8$ **22** $-x^2+7x$

23 $6a^2+10ab$ **24** $10m-5m^2$ **25** $7x+10$ **26** $7a-11$

27 x^2+5x-8 **28** $10-5y$ **29** $5x$ ↻ L1 **30** $4p+5q$

31 \$45 and \$75 ↻ Ratios **32** $2x+3$ **33** $5x^2-7x$

34 $7(x+4)=7x+28$ (distributive law)

35 $-3\times(-5)=+15$, so $-3(x-5)=-3x+15$

36 a) $3x+4(x-8)=7x-32$ b) \$108

G Gold round

37 $x^2+9x+14$ **38** $a^2+2a-24$ **39** $2m^2-5m-3$ **40** $x^2+10x+25$

41 y^2-9 **42** $9a^2-12a+4$ **43** x^2-16 **44** $6x^2+7x-3$

45 2

46 a) $n^2+2n+1-n^2=2n+1$ b) $n=100$: 201

47 $(x+4)(x+1)-x^2=x^2+5x+4-x^2=5x+4$

3 Factorising

B Bronze round

1 6 **2** 4 **3** $3a$ **4** x

5 $3(x+4)$ **6** $5(a-4)$ **7** $2(2m+3)$ **8** $7(y-2)$

9 $8x-12$ ↻ L2 **10** $x(x+3)$ **11** $a(b+5)$ **12** $5(2-3k)$

13 5 ↻ Integers **14** $9(p+1)$ **15** $12a^2$ ↻ L1 **16** $2x(3y+4)$

17 $x=5$ ↻ Equations **18** $4(3a-2b)$

S Silver round

19 $x(x-7)$ **20** $2a(2a+5)$ **21** $3m(3n-2)$ **22** $-4(x+3)$

23 $-5(2a-3)$ **24** $-x(x+6)$ **25** $5pq(3p+4q)$ **26** $3(x^2+2x+3)$

27 $2a(a-2b+3)$ **28** $-6+15y$ ↻ L2 **29** \$68 ↻ Percentages **30** $6xy(3x-2)$

31 14 ↻ L1 **32** Yes: $3(2)+1=7$ ↻ Linear

33 $x+2$ **34** $2a-3$ **35** $x+2$

36 a) $2x(6x-4)=12x^2-8x$ ✓ b) $6x-4$ still has a common factor of 2; $4x(3x-2)$

37 $(2x+5)$ cm

38 $17(23+77) = 17 \times 100 = 1700$

G Gold round

39 $(x+3)(a+b)$ **40** $(y-2)(x-4)$ **41** $(x-5)(x+5)$ **42** $(2m-3)(2m+3)$

43 $(x+3)(x+4)$ **44** $(x-5)(x+3)$ **45** $(x+4)(x-3)$ **46** $(3a-1)(3a+1)$

47 $2(x-3)(x+3)$

48 $(53-47)(53+47) = 6 \times 100 = 600$

49 $n(n+1)$: the product of two consecutive whole numbers, one of which must be even

4 Indices

B Bronze round

1 5^4 **2** 32 **3** 10 000 **4** 25

5 1 **6** $20xy$ L1 **7** 9 **8** 4

9 $4(2x-5)$ L3 **10** 2^5 $(32 > 25)$ **11** $10a+15$ L2 **12** $\frac{2}{3}$ Fractions

13 2^7 **14** 5^6 **15** 3^8 **16** 6^4

17 $2^3 \times 3$ **18** $2^2 \times 5^2$ **19** $2 \times 3^2 \times 5$ **20** $2^3 \times 7$

S Silver round

21 49 **22** 4 **23** 9 **24** 1000

25 21 **26** 24 **27** 72 **28** $-3(2a+3)$ L3

29 $2^2 \times 3^3 \times 5$ **30** $2^2 \times 3^2 \times 5 \times 7$ **31** 42 **32** 13 L1

33 25% Percentages **34** 7 and 8 **35** $n = 7$ **36** $n = 7$

37 $x^2 - x$ L2 **38** 1

39 False: $2^3 \times 2^4 = 2^7 = 128$, but $4^7 = 16\,384$. Add the indices; do not multiply the bases.

40 a) $2^{10} = 1024$ b) 20 hours ($2^{19} = 524\,288$, $2^{20} = 1\,048\,576$)

41 Index law: $5^{3-3} = 5^0$. Dividing a number by itself: $125 \div 125 = 1$. So $5^0 = 1$.

G Gold round

42 a^9 **43** x^5 **44** m^{12} **45** $12x^7$

46 $4a^6$ **47** $3p^3q$ **48** 6 **49** $\frac{1}{8}$

50 $27x^6y^3$ **51** $2a$ **52** x **53** $\frac{1}{100}$

54 $2^{30} = (2^3)^{10} = 8^{10}$ and $3^{20} = (3^2)^{10} = 9^{10}$, so 3^{20} is larger.

55 $4^n = 2^{2n}$, so $2n = 10$ and $n = 5$

5 Square roots & Pythagoras' theorem

B Bronze round

1 7 **2** 12 **3** 13 **4** 12

5 3 **6** 32 L4 **7** $6x$ L1 **8** $5(y+3)$ L3

9 5 and 6 **10** 5.5 ($5.4^2 = 29.16$, $5.5^2 = 30.25$)

11 10 cm **12** 15 cm **13** 17 cm **14** $\sqrt{74}$ cm

15 35% Percentages **16** PR

S Silver round

17 12 cm **18** 24 cm **19** $\sqrt{84}$ cm **20** 10 cm

21 $\sqrt{13}$ **22** Yes: $81 + 1600 = 1681 = 41^2$ **23** No: $36 + 81 = 117 \neq 121$ **24** Yes: $400 + 441 = 841 = 29^2$

25 $-6x + 2$ L2 **26** 17 L4 **27** 6 L1 **28** $x(x+4)$ L3

29 9.5 (9.49) **30** 5 : 3 Ratios

31 $\sqrt{25 - 1.96} = \sqrt{23.04} = 4.8$ m

32 $\sqrt{5.76 + 1} = \sqrt{6.76} = 2.6$ m

33 Direct: $\sqrt{360\,000 + 62\,500} = 650$ m; she walked 850 m, so 200 m shorter.

34 e.g. 6, 8, 10 and 9, 12, 15. $(3k)^2 + (4k)^2 = 9k^2 + 16k^2 = 25k^2 = (5k)^2$.

G Gold round

35 $2\sqrt{3}$ **36** $5\sqrt{2}$ **37** $6\sqrt{2}$ **38** $7\sqrt{3}$

39 $6\sqrt{10}$ **40** $\sqrt{5}$ **41** $\sqrt{80} = 4\sqrt{5}$ **42** 12

43 4 **44** $3 + \sqrt{3}$ **45** 4 **46** $3\sqrt{2}$

47 run 3, rise 4: $\sqrt{9+16} = 5$ units

48 $(\sqrt{5})^2 + (\sqrt{20})^2 = 5 + 20 = 25 = 5^2$ ✓

49 Base diagonal $\sqrt{144+16} = \sqrt{160}$; rod $= \sqrt{160+9} = \sqrt{169} = 13$ cm

6 Equations & formulas

B Bronze round

1 $x = 6$ **2** $y = 1$ **3** $m = 7$ **4** $k = 15$

5 $x = 4$ **6** $a = 7$ **7** $n = 12$ **8** $6x - 12$ ↻ L2

9 $x = 6$ **10** $p = -3$ **11** 243 ↻ L4 **12** $x = \pm 6$

13 5 ↻ L5 **14** $x = 4$ **15** $2(3a + 4)$ ↻ L3 **16** $x = 4$

17 $x = 11$ **18** 30 ↻ Percentages

S Silver round

19 $x = 4$ **20** $a = -4$ **21** $x = 12$ **22** $x = 3$

23 $x = -4$ **24** $x = 10$ **25** $x = 6$ **26** $x = 19$

27 $x = \pm 4$ **28** $x = \pm 5$ **29** $x = \pm\sqrt{8}$ **30** 25 ↻ L4

31 $x + 11$ ↻ L2 **32** Yes: $64 + 225 = 289$ ↻ L5 **33** 37.5% ↻ Percentages **34** $\frac{7}{12}$ ↻ Fractions

35 $t = 5$ **36** $h = 7.5$ **37** $v = 5$ **38** $r = 7$

39 $4n - 7 = 2n + 9$, so $2n = 16$ and $n = 8$

40 $4.50 + 2.20d = 37.50$, so $2.20d = 33$ and $d = 15$ km

G Gold round

41 $m = 4$ **42** $x = -11$ **43** $x = 9$ **44** $x = 12$

45 $x = 7$ **46** $y = 5$ **47** $x = 7$ **48** $x = 4$

49 $x = 3$

50 $x = \dfrac{y-2}{3}$ **51** $r = \dfrac{C}{2\pi}$ **52** $a = \dfrac{v^2 - u^2}{2s}$

53 $2(2w + 3) + 2w = 42 \Rightarrow 6w + 6 = 42 \Rightarrow w = 6$; 6 cm by 15 cm

54 $n + (n+2) + (n+4) = 138 \Rightarrow 3n = 132$; 44, 46, 48

7 Area & volume

B Bronze round

1 36 m² **2** 49 cm² **3** 30 cm² **4** 60 cm²

5 30 cm² **6** 30 cm² **7** 42 m² **8** $x = 5$ ↻ L6

9 64 ↻ L4 **10** 60 cm³ **11** 216 cm³ **12** 120 cm³

13 10 ↻ L5 **14** $x^2 + 5x$ ↻ L2 **15** $3(3m - 4)$ ↻ L3 **16** 10π cm³

17 90 cm³ **18** 1.8 ↻ Percentages

S Silver round

19 $80 - 18 = 62$ cm² **20** $60 + 24 = 84$ cm² **21** $A = \frac{3}{2}(4 + 10) = 21$; $V = 420$ cm³

22 60 cm² **23** 8 cm ↻ L6 **24** 9 cm ↻ L6 **25** 200π cm³

26 300π cm³ **27** 310 cm³ **28** $h = 12$; 60 cm² ↻ L5 **29** $6x^2$ ↻ L1

30 5 cm **31** 375 m³ **32** $4x + 12$ ↻ L2 **33** 7 cm ↻ L4

34 Two identical copies of the triangle fit together to make a parallelogram with base b and height h, so the triangle is $\frac{1}{2}bh$.

35 $A = \frac{2}{2}(3.5 + 5.5) = 9$ m²; $V = 9 \times 0.3 = 2.7$ m³

G Gold round

36 $\pi(5^2 - 4^2) \times 100 = 900\pi$ cm³

37 $36\pi h = 360\pi$, so $h = 10$ cm

38 a) $x^2(x + 3) = x^3 + 3x^2$ b) $64 + 48 = 112$ cm³

39 $\frac{5}{2}(2x + 4) = 45 \Rightarrow 2x + 4 = 18 \Rightarrow x = 7$

40 $1000 - \pi \times 2^2 \times 10 = (1000 - 40\pi)$ cm³

41 a) $\pi(2r)^2 h = 4\pi r^2 h$: ×4 b) $\pi(2r)^2(2h) = 8\pi r^2 h$: ×8

Pre-test and post-test – marking guide (20 marks each)

Q	PRE-TEST ANSWER	POST-TEST ANSWER	MARK

1a	$3x + 3y$	$4m + 2n$	1
1b	$12a^2b$	$10x^2y$	1
1c	14	7	1
2a	$x^2 - 4x$	$y^2 + 5y$	1
2b	$-6x + 15$	$-12x + 8$	1
2c	$2a + 14$	$a + 14$	1
3a	$6(2m + 3)$	$5(3k + 4)$	1
3b	$3x(2x - 3y)$	$4a(2a - 3b)$	1
3c	$-2(2a + 5)$	$-3(2x + 5)$	1
4a	$5^3 = 125$	$3^3 = 27$	1
4b	4	25	1
4c	$2^2 \times 3^2 \times 5$	$2^2 \times 3^2 \times 7$	1
5a	15 cm	25 cm	1
5b	$\sqrt{75}$ cm (accept $5\sqrt{3}$)	$\sqrt{48}$ cm (accept $4\sqrt{3}$)	1
5c	No: $25 + 121 = 146 \neq 144$	Yes: $64 + 225 = 289 = 17^2$	1
6a	$x = 5$	$x = 7$	1
6b	$a = -4$	$a = 4$	1
6c	$x = \pm 6$	$x = \pm 5$	1
7a	28 cm^2	40 cm^2	1
7b	45π cm^3	28π cm^3	1

Award the mark for the correct final answer; accept equivalent forms (e.g. $5\sqrt{3}$ for $\sqrt{75}$). In Q3 a correct but not fully factorised answer scores 0, because "factorise fully" is the skill tested. Q2b and Q3c use negative coefficients (★, Stage 5 A in the syllabus but taught in NCM 8 Advanced).